HW 6

1) a) $\theta = \{x/A, y/A, z/B\}$

b) no general unifier

c) $\theta = \{x/B, y/A\}$

d) $y \to x \quad x \to John$

$\theta = \{x/John, y/John\}$

e) no general unifier

2) a) 1. $(\forall f)(Food(f) \Rightarrow Likes(John, f))$

2. Food(Apples)

3. Food(Chicken)

4. $(\forall f)(\forall h)[(Eats(h, f) \& \sim Killed(h, f)) \Rightarrow Food(f)]$

5. $(\forall y)[\forall t)[Killed(y, t) \Rightarrow \sim Alive(y)]]$ (ELI)

6. Eats(Bill, peanuts) & Alive(Bill)

7. $(\forall f)(Eats(Bill, f) \Rightarrow Eats(Sue, f))$

b) 1. $(\forall f)(\sim Food(f) \mid Likes(John, f))$

$\sim Food(f) \mid Likes(John, f)$

2. Food(Apples)

3. Food(Chicken)

4. $(\forall f)(\forall h)[\sim(Eats(h, f) \& \sim Killed(h, f)) \mid Food(f)]$

~~$(\forall f)[(\forall h)(\sim Eats(h, f) \mid Killed(h, f)) \mid Food(f)]$~~

$(\forall f)(\forall h)(\sim Eats(h, f) \mid Killed(h, f) \mid Food(f))$

$\sim Eats(h, f) \mid Killed(h, f) \mid Food(f)$

5. $(\forall y)(\forall t)[\sim Killed(y, t) \mid \sim Alive(y)]$

~~$(\forall y)[(\forall t)(\sim Killed(y, t)) \mid \sim Alive(y)]$~~

$(\forall y)(\forall t)[\sim Killed(y, t) \mid \sim Alive(y)]$

$\sim Killed(y, t) \mid \sim Alive(y)$

6. Eats(Bill, Peanuts) & Alive(Bill)

Eats(Bill, peanuts)

Alive(Bill)

7. $(\forall f)(Eats(Bill, f) \Rightarrow Eats(Sue, f))$

$(\forall f)(\sim Eats(Bill, f) \mid Eats(Sue, f))$

$\sim Eats(Bill, f) \mid Eats(Sue, f)$

c) Prove John likes peanuts using resolution.

KB, Δ:

1. $\sim Food(f) \mid Likes(John, f)$
2. $Food(Apples)$
3. $Food(Chicken)$
4. $\sim Eats(h, f) \mid Killed(h, f) \mid Food(f)$
5. $\sim Killed(y, f) \mid \sim Alive(y)$
6. $Eats(Bill, peanuts)$
7. $Alive(Bill)$
8. $\sim Eats(Bill, f) \mid Eats(Sue, f)$
9. $\sim Likes(John, peanuts)$
10. $Killed(Bill, Peanuts) \mid Food(peanuts)$ by 4,6 $\{h/Bill, f/peanuts\}$
11. $\sim Killed(Bill, f)$ by 5,7 $\{y/Bill\}$
12. $Food(peanuts)$ by 9,10 $\{f/peanuts\}$
13. $Likes(John, peanuts)$ by 1,11 $\{f/peanuts\}$
14. $\{\}$ by 9,12

Since we have a contradiction, John likes peanuts.

d) What does Sue eat?

1. ~Food(f) | Likes(John, f)
2. Food(Apples)
3. Food(chicken)
4. ~Eats(h, f) | Killed(h, f) | Food(f)
5. ~Killed(y, t) | ~Alive(y)
6. Eats(Bill, Peanuts)
7. Alive(Bill)
8. ~Eats(Bill, f) | Eats(Sue, f)
9. ~Eats(SUE, peanuts) negation
10. Eats(Sue, peanuts) by 6, 8 {P/peanuts}
11. {} Empty by 9, 9 {f/peanuts}

Sue eats peanuts.

e)

1. ~Food(f) | Likes(John, f)
2. Food(Apples)
3. Food(chicken)
4. ~Eats(h, f) | Killed(h, f) | Food(f)
5. ~Killed(y, t) | ~Alive(y)
6. ~Eats(Bill, f) | Eats(Sue, f)
7. Eats(h, f) | Die(h)
8. ~Die(h) | ~Alive(h)
9. Alive(Bill)
10. ~Eats(Sue, Apples)
11. ~Eats(Bill, Apples)
12. ~Die(Bill)
13. ~Alive(Bill)
14. {}

New:

a. (Af)(Ah)[~Eats(h, f) ⇒ Die(h)]
   Eats(h, f) | Die(h)
b. (A h)[Die(h) ⇒ ~Alive(h)]
   ~Die(h) | ~Alive(h)
c. Alive(Bill)

Negation

by 6, 10 {P/Apples}
by 7, 11 {h/Bill, P/Apples}
by 8, 12 {h/Bill}
by 9, 13

So, Sue eats apples.

c) Attempt 2:

1. ~ Food(f) | Likes(John, f)
2. Food(Apples)
3. Food(chicken)
4. ~ Eats(h, f) | Killed(h, f) | Food(f)
5. ~ Killed(y, t) | ~ Alive(y)
6. ~ Eats(Bill, f) | Eats(Sue, f)
7. Eats(h, f) | Die(h)
8. ~ Die(h) | ~ Alive(h)
9. Alive(Bill)
10. ~ Eats(Sue, chicken) negation
11. ~ Eats(Bill, chicken) by 6, 10 {f/chicken}
12. Die(Bill) by 7, 11 {h/Bill}
13. ~ Alive(Bill) by 8, 12 {h/Bill}
14. {} contradiction by 9, 13

Sue eats chicken.